

# Unit 101 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 97.4 ac, BAF 20 variable-radius cruise on a 300 ft grid, 45 plots, 478 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

**Table 1. Stand summary from the BAF 20 cruise, per acre**

| STAND DENSITY        |                                       | STAND INFO                  |                                                           |
|----------------------|---------------------------------------|-----------------------------|-----------------------------------------------------------|
| Basal area           | 212 sq ft/ac (1 SE of mean, BA 1.9 %) | Type                        | Sierran mixed conifer                                     |
| Trees per acre       | 123                                   | CWHR size / density         | 4 / D (cover 83 %)                                        |
| QMD                  | 17.8 in                               | SDI                         | 288 (51 % of max 567)                                     |
| SAWTIMBER (>= 10 in) |                                       | BIOMASS (< 10 in)           |                                                           |
| BA   QMD             | 203 sq ft/ac   19.8 in                | BA   QMD                    | 9 sq ft/ac   7.6 in                                       |
| TPA                  | 95                                    | TPA                         | 28                                                        |
| Volume               | 29.8 MBF/ac (5943 cu ft)              | Volume                      | 2.9 green tons/ac (130 cu ft)                             |
| TREATMENT TARGET     |                                       | CRUISE DESIGN               |                                                           |
| Leave                | 35 % of max SDI = ~146 sq ft/ac BA    | Sampling error (volume)     | 10.4 % at 95 % (CV 35 %, t = 2.02, n = 45)                |
| Species (BA)         | PP 70, WF 60, IC 32, DF 29, SP 21     | Stratum standard            | 40 % of volume - MEETS; plots for standard: 4             |
|                      |                                       | Sampling error (basal area) | 3.9 % at 95 % (CV 13 %); secondary, not the standard      |
|                      |                                       | Plot count                  | 45 vs 20-plot local practice, not a handbook standard: ok |

Cubic volume = total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium) net of defect; Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006); green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber, so applied to total-stem CVTS it overstates sawlog volume; merchantable cubic volume would be the consistent pairing. Sampling error at 95 % confidence with Student's t on plot net cubic volume per acre, the quantity the handbook standards apply to (FSH 2409.12 ch. 40, 41.1(5)(a) and (b)); stratum standard 40 % of volume for tree-measurement sales; the basal-area error is secondary. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables exclude records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise\_data.xlsx, Standards and assumptions.

**Table 2. Stand table by 4-inch DBH class, per acre**

| DBH class, in | Trees      | TPA          | BA           | cu ft       | MBF         |
|---------------|------------|--------------|--------------|-------------|-------------|
| 4-7.9         | 8          | 15.6         | 3.6          | 51          | 0.3         |
| 8-11.9        | 34         | 27.7         | 15.1         | 241         | 1.2         |
| 12-15.9       | 74         | 30.7         | 32.9         | 670         | 3.4         |
| 16-19.9       | 82         | 21.2         | 36.4         | 870         | 4.4         |
| 20-23.9       | 68         | 11.7         | 30.2         | 831         | 4.2         |
| 24-27.9       | 50         | 6.0          | 22.2         | 686         | 3.4         |
| 28-31.9       | 52         | 4.8          | 23.1         | 782         | 3.9         |
| 32-35.9       | 33         | 2.3          | 14.7         | 534         | 2.7         |
| 36-39.9       | 25         | 1.4          | 11.1         | 439         | 2.2         |
| 40-43.9       | 18         | 0.9          | 8.0          | 310         | 1.6         |
| 44-47.9       | 10         | 0.4          | 4.4          | 173         | 0.9         |
| 48-51.9       | 3          | 0.1          | 1.3          | 59          | 0.3         |
| 52-55.9       | 9          | 0.3          | 4.0          | 189         | 0.9         |
| 56-59.9       | 4          | 0.1          | 1.8          | 88          | 0.4         |
| 60-63.9       | 7          | 0.2          | 3.1          | 151         | 0.8         |
| <b>All</b>    | <b>477</b> | <b>123.4</b> | <b>212.4</b> | <b>6073</b> | <b>30.5</b> |

**Table 3. Stock table by species, per acre**

| Species    | Trees      | TPA          | BA           | QMD         | cu ft       | MBF         |
|------------|------------|--------------|--------------|-------------|-------------|-------------|
| PP         | 158        | 45.2         | 70.2         | 16.9        | 2158        | 10.8        |
| WF         | 134        | 31.5         | 59.6         | 18.6        | 1701        | 8.5         |
| IC         | 71         | 19.1         | 31.6         | 17.4        | 768         | 3.9         |
| DF         | 66         | 14.3         | 29.3         | 19.4        | 849         | 4.3         |
| SP         | 47         | 13.3         | 20.9         | 17.0        | 574         | 2.9         |
| PPP        | 1          | 0.1          | 0.4          | 35.1        | 23          | 0.1         |
| <b>All</b> | <b>477</b> | <b>123.4</b> | <b>212.4</b> | <b>17.8</b> | <b>6073</b> | <b>30.5</b> |

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 101: plots on the 300 ft grid (orange = QA flag)

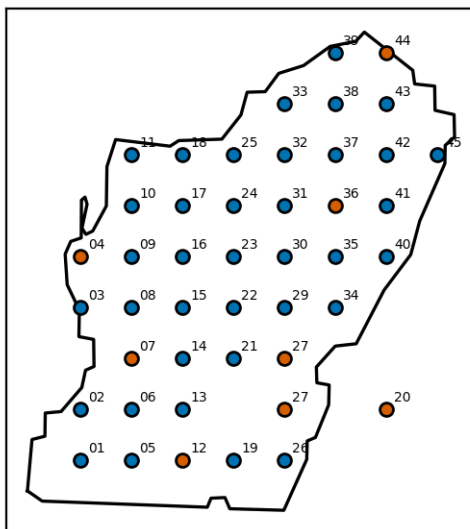


Figure 1. Plot locations with QA status.

**QA findings: 7 item(s) need crew follow-up**

| Plot   | Finding                                                                  |
|--------|--------------------------------------------------------------------------|
| 101-04 | tree 1: unknown species code 'PPP'                                       |
| 101-07 | tree 9: height 32 ft implausible for DBH 27.8                            |
| 101-12 | tree 1: DBH 999.0 out of range                                           |
| 101-20 | plot location falls outside its unit boundary (GPS or unit number error) |
| 101-27 | duplicate plot id recorded 2 times                                       |
| 101-36 | tree 2: height missing                                                   |
| 101-44 | tree 1: height 15 ft implausible for DBH 34.0                            |

Table 4. Plot list

| Plot   | Trees in | BA sq ft/ac | Slope % | Status |
|--------|----------|-------------|---------|--------|
| 101-01 | 12       | 240         | 11      | ok     |
| 101-02 | 10       | 200         | 14      | ok     |
| 101-03 | 6        | 120         | 28      | ok     |
| 101-04 | 9        | 180         | 25      | FLAG   |
| 101-05 | 9        | 180         | 16      | ok     |
| 101-06 | 12       | 240         | 14      | ok     |
| 101-07 | 13       | 260         | 9       | FLAG   |
| 101-08 | 11       | 220         | 10      | ok     |
| 101-09 | 11       | 220         | 19      | ok     |
| 101-10 | 12       | 240         | 25      | ok     |
| 101-11 | 9        | 180         | 24      | ok     |
| 101-12 | 10       | 200         | 8       | FLAG   |
| 101-13 | 11       | 220         | 5       | ok     |
| 101-14 | 10       | 200         | 7       | ok     |
| 101-15 | 10       | 200         | 3       | ok     |
| 101-16 | 11       | 220         | 6       | ok     |
| 101-17 | 11       | 220         | 4       | ok     |
| 101-18 | 10       | 200         | 10      | ok     |
| 101-19 | 10       | 200         | 8       | ok     |
| 101-20 | 11       | 220         | 9       | FLAG   |
| 101-21 | 11       | 220         | 10      | ok     |
| 101-22 | 10       | 200         | 5       | ok     |
| 101-23 | 11       | 220         | 3       | ok     |
| 101-24 | 11       | 220         | 5       | ok     |
| 101-25 | 11       | 220         | 6       | ok     |
| 101-26 | 15       | 300         | 15      | ok     |
| 101-27 | 11       | 220         | 9       | FLAG   |
| 101-27 | 9        | 180         | 17      | FLAG   |
| 101-29 | 12       | 240         | 8       | ok     |
| 101-30 | 11       | 220         | 3       | ok     |
| 101-31 | 11       | 220         | 3       | ok     |
| 101-32 | 11       | 220         | 4       | ok     |
| 101-33 | 9        | 180         | 7       | ok     |
| 101-34 | 9        | 180         | 9       | ok     |
| 101-35 | 10       | 200         | 2       | ok     |
| 101-36 | 12       | 240         | 3       | FLAG   |
| 101-37 | 11       | 220         | 6       | ok     |
| 101-38 | 11       | 220         | 14      | ok     |
| 101-39 | 10       | 200         | 4       | ok     |
| 101-40 | 12       | 240         | 1       | ok     |
| 101-41 | 12       | 240         | 2       | ok     |
| 101-42 | 10       | 200         | 4       | ok     |
| 101-43 | 9        | 180         | 8       | ok     |
| 101-44 | 11       | 220         | 6       | FLAG   |
| 101-45 | 10       | 200         | 10      | ok     |